An almost-entropic Ingleton score below −0.09738

Conditional hashing of a binary parity structure

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Abstract

We prove an improved lower bound of 0.097387436922 on the largest normalised Ingleton violation among four-variable almost-entropic vectors. Previously reported constructions reach approximately 0.0925, used here as a rounded benchmark; the best published universal upper bound is $19/124 \approx 0.15322581$. The main novelty is a construction combining local parity labels, one common binary linear hash, simultaneous conditional privacy amplification and joint conditioning. Applied to a 24-atom source of Matúš and Csirmaz, it produces an auxiliary determined by any three observations, with one hashed degree of freedom retaining asymptotically full conditional entropy given any pair's complete records. Conditioning subtracts a multiple of the uniform-matroid rank function $U_{3,4}$ from all fifteen entropy coordinates. The Ingleton expression is preserved while the joint entropy decreases, improving the normalised violation. We give a general six-test criterion for this subtraction. The main theorem has been formalised and kernel-checked in Lean, including the closed rate interval, the score enclosure and the existence of finite-alphabet distributions with signed score below −0.09738.

Index terms—Entropy region, Ingleton inequality, privacy amplification, formal verification.

I Introduction

Four finite random variables determine fifteen joint entropies, $h(S) = H(X_S)$ for the nonempty subsets of $N = \{A, B, C, D\}$. Which such vectors arise from distributions, or can be approached arbitrarily closely? This is the four-variable entropy-region problem, with applications to network coding, secret sharing and information inequalities [1, 2]. We put $h(\varnothing) = 0$, measure entropy in bits, and write $AB = (X_A, X_B)$ for joint observations.

The closure of the finite-alphabet entropy region Γ_4^* is the *almost-entropic cone* $\overline{\Gamma_4^*}$. It is a closed convex cone [3] between two explicit cones:

$$\mathcal{L}_4 \subsetneq \overline{\Gamma_4^*} \subsetneq \Gamma_4.$$

Here the Shannon cone Γ_4 consists of monotone submodular set functions vanishing at the empty set. The linear cone $\mathcal{L}_4$ is generated, with closure, by entropy vectors of linear observations of uniform finite-field randomness. In four variables it is exactly the part of Γ_4 satisfying the six Ingleton inequalities [4, 5]. For at most three variables, Shannon's inequalities describe the whole closure [3]. For four, Zhang and Yeung showed that further inequalities are necessary [6], and Matúš proved that no finite collection of linear inequalities suffices [7]. The geometry of the intervening almost-entropic boundary remains unknown. The distinction between entropic and almost-entropic points also affects conditional information inequalities [8].

A The Ingleton gap

We use the following instance of Ingleton's inequality [9]:

$$\mathrm{J}(h) = I(C; D \mid A) + I(C; D \mid B) + I(A; B) - I(C; D) \geq 0. \tag{1}$$

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There are six distinct instances, one for each choice of the pair in the final mutual-information term. Linear observations satisfy all six; general distributions may violate them. For $h(N) > 0$, following [1, Definition 3], write the signed score as $\sigma(h) = \mathrm{J}(h)/h(N)$ and the normalised violation as $\nu(h) = -\sigma(h)$, and set

$$\nu^* = \sup_{f \in \overline{\Gamma_4^*},\ f(N)>0} \nu(f). \tag{2}$$

On the normalised slice $f(N) = 1$, $\nu = -\mathrm{J}$ is a linear height beyond the chosen Ingleton hyperplane, in the units fixed by (1). It is invariant under positive scaling; a Euclidean distance would also require a choice of norm. The same supremum holds over finite entropic vectors, by density and continuity where $f(N) > 0$. A construction gives a lower bound on ν^*; a universal inequality gives an upper bound. The Shannon cone permits 1/4 [1], while Chang's recent preprint proves the upper bound 19/124 for the almost-entropic cone [10].

B Earlier constructions and the new bound

The four-atom conjecture of Dougherty, Freiling and Zeger proposed an optimum of approximately 0.089373 [1]. Matúš and Csirmaz refuted it using a 40-atom source, tightening and almost-entropy-preserving maps [2, Example 2, Theorem 4]. Boston and Nan subsequently optimised the associated transformed violation [11].

Construction	Method/status	Positive violation ν
Four-atom source [1]	numerical ordinary score	≈ 0.089373
Matúš–Csirmaz [2]	transformed limiting score	≈ 0.09243
Boston–Nan [11]	optimised transformed score	≈ 0.0925000777
This paper	rigorous almost-entropic limit	[0.097387436922, 0.097387436923]

In our orientation the transformed violation is

$$\widetilde{\nu}(h) = \frac{-\mathrm{J}(h)}{h(ACD) + h(BCD) + h(AB) - 2h(N)}.$$

For a tight source, meaning that every triple has entropy $h(N)$, its denominator is $h(AB)$. Matúš–Csirmaz's theorem shows that such transformed violations are limits of ordinary violations [11, Theorem 2.6]. The 2024 numerical value 0.0925001031 is reported with the tiny change attributed to numerical stability [12, Section 5.2, Table 2]. To the best of our knowledge, the strongest previously reported constructions reach approximately 0.0925, which we use as a rounded benchmark. Our construction and the best published universal bound leave

$$0.097387436922 \le \nu^* \le \frac{19}{124} = 0.15322580645161\ldots\ . \tag{3}$$

C The result and its organising idea

Our source is the $s = t = 0$ specialisation of Matúš–Csirmaz's Example 2. Choose one of the three columns in Table 2 with probabilities $(P, Q, T) = (0.74830195, 0.22122546, 0.03047259)$, then choose a row uniformly and read its four symbols as the observations. Let h be its entropy vector, let $\mathcal{H} = H(P, Q, T)$ be the entropy of the column choice, and write h_2 for binary entropy. Put $u(S) = \min\{3, |S|\}$, the rank function of the uniform matroid $U_{3,4}$, and $b = I(C; D \mid AB)$.

Theorem 1. *For every $0 \le R \le b$, $h - Ru \in \overline{\Gamma_4^*}$. For $z = h - bu$, $z(N) > 0$ and its exact signed score is*

$$\frac{\mathrm{J}(z)}{z(N)} = \frac{2 - 2P + 2h_2(P) + h_2(T) - 2\mathcal{H} - h_2(Q)}{3 - 2\mathcal{H} + 3h_2(Q)}.$$

It satisfies

$$-0.097387436923 \;\le\; \frac{\mathrm{J}(z)}{z(N)} \;\le\; -0.097387436922. \tag{4}$$

Consequently $\nu^* \ge 0.097387436922$, and some four finite-alphabet random variables have signed Ingleton score below -0.09738.

The idea is to construct an auxiliary with a prescribed information profile. Conditioning on a finite auxiliary W changes $h(S)$ to

$$H(X_S \mid W) = h(S) - I(X_S; W).$$

Thus, on n independent source copies, subtracting Ru calls for $I(X_S^n; W)/n \to R\min\{3, |S|\}$. We seek a three-key auxiliary that any three observations determine, while two observations leave one key's worth of entropy and one observation leaves two.

Each observer computes a local bit Y_i and the four bits sum to zero. One common binary linear map hashes their length-n sequences to k-bit keys, preserving this parity exactly. A pair already knows its two keys. It is missing one further hashed binary form that completes them to three independent coefficient rows. The six privacy requirements correspond to the pairs AB, AC, AD, BC, BD, CD: for each pair, the missing hashed form must retain $k - O(1)$ bits of conditional entropy even after both observers' *complete records* are revealed. A single matrix must meet all six requirements. These requirements allow dependent, nonuniform labels and imply the four singleton requirements used in the proof. The resulting information accounting is

Observed subset size	1	2	3 or 4
$H(W \mid X_S^n)$	$2k + O(1)$	$k + O(1)$	0
$I(X_S^n; W)$	$k + O(1)$	$2k + O(1)$	$3k + O(1)$

The constants are independent of n. Since $\mathrm{J}(u) = 0$ and $u(N) = 3$, the subtraction preserves the Ingleton numerator and lowers the joint entropy by $3R$.

Tightening and convolution already provide important ways of modifying entropy vectors [13, 2], and privacy amplification supplies the hashing tool [14, 15, 16]. To our knowledge, the new ingredient here is their coordinated use to realise the non-modular subtraction Ru from a source with local parity labels. For the same source, the ordinary, Matúš–Csirmaz-transformed and new violations are approximately 0.08365, 0.08778 and 0.09739, respectively. The main theorem is kernel-checked in Lean; its exact formal scope is stated in Section VI.

Section II proves the general six-test criterion. Section III verifies its hypotheses for the source; Section IV computes the score and the geometry of the subtraction path; Section V gives finite conditional laws and a sampling recipe. The hashing proof and the numerical enclosure are supplied in the appendices.

II Parity subtraction from six privacy tests

The construction rests on a distinction between an exact algebraic relation and the conditional entropy available for hashing. Four local bits satisfying one parity relation have three algebraic degrees of freedom. A common linear hash preserves that relation, while privacy amplification preserves asymptotically one bit of conditional entropy per output bit of each missing form. The following theorem gives a sufficient condition for turning this structure into entropy subtraction. All bit arithmetic below is over $\mathbb{F}_2$, with addition modulo two.

Theorem 2 (Parity-subtraction criterion)**.** *Let X_A, X_B, X_C, X_D be finite random variables with entropy vector h. Suppose deterministic local labels $Y_i = \chi_i(X_i) \in \mathbb{F}_2$ satisfy $Y_A + Y_B + Y_C + Y_D = 0$ almost surely. Put $U = Y_A$, $V = Y_B$, $E = Y_A + Y_C$, and let*

$$c = \min\{H(E \mid X_A X_B),\ H(V \mid X_A X_C),\ H(V \mid X_A X_D),\\ H(U \mid X_B X_C),\ H(U \mid X_B X_D),\ H(U \mid X_C X_D)\}.$$

Then, with $u(S) = \min\{3, |S|\}$,

$$h - Ru \in \overline{\Gamma_4^*} \qquad (0 \le R \le c).$$

No uniformity or independence of the labels is assumed.

Each quantity defining c is a privacy budget: how much randomness remains in a missing bit after a pair's complete observations are revealed. The minimum is the rate a single hash can achieve in all six situations. In particular the criterion concerns the finite source and its local labels, rather than its entropy vector alone.

A Algebraic completion and singleton privacy

The parity identity writes the four labels as

$$(Y_A, Y_B, Y_C, Y_D) = (U, V, U + E, V + E). \tag{5}$$

Their coefficient vectors in the formal basis (U, V, E) are $(1,0,0), (0,1,0), (1,0,1), (0,1,1)$. Any two have rank two, and any three have rank three. These are statements about the linear maps, even when the random inputs are dependent. For every pair, Table 1 gives a missing form L_{ij} and the explicit completion after hashing. Any other completing binary linear form differs by a form known to the pair, so its conditional entropy given $X_i X_j$ is the same.

Pair	L_{ij}	(K_A, K_B, K_C) from the pair's keys and $N_{ij} = ML_{ij}^n$
AB	E	$(K_A, K_B, K_A + N_{ij})$
AC	V	(K_A, N_{ij}, K_C)
AD	V	$(K_A, N_{ij}, K_A + K_D + N_{ij})$
BC	U	(N_{ij}, K_B, K_C)
BD	U	$(N_{ij}, K_B, N_{ij} + K_B + K_D)$
CD	U	$(N_{ij}, N_{ij} + K_C + K_D, K_C)$

Table 1: The six basis completions. Here $K_i = MY_i^n$ and $K_D = K_A + K_B + K_C$. Conversely, $N_{AB} = K_A + K_C$, $N_{AC} = N_{AD} = K_B$, and $N_{BC} = N_{BD} = N_{CD} = K_A$. These are invertible changes of coordinates, with no distributional assumption.

The six pair tests also guarantee the four singleton tests needed in the proof. Conditioning reduces entropy, and (5) gives

$$\begin{aligned} &H(Y_B \mid X_A) \ge H(V \mid X_A X_C) \ge c,\\ &H(Y_A \mid X_B) \ge H(U \mid X_B X_C) \ge c,\\ &H(Y_D \mid X_C) \ge H(Y_D \mid X_A X_C) = H(V \mid X_A X_C) \ge c,\\ &H(Y_C \mid X_D) \ge H(Y_C \mid X_B X_D) = H(U \mid X_B X_D) \ge c. \end{aligned} \tag{6}$$

For the third line, (X_A, X_C) fixes E, so $Y_D = V + E$ has the same conditional entropy as V. The fourth line uses $Y_C = U + Y_B + Y_D$, with Y_B, Y_D fixed by (X_B, X_D). All side information here consists of the full observed variables.

B One linear hash for every test

We use the following finite-alphabet form of conditional privacy amplification [14, 15, 16]. Its elementary proof is included in Appendix A.

Lemma 3 (Simultaneous linear hashing)**.** *Let (Z_ℓ, G_ℓ), $1 \le \ell \le q$, be a nonempty finite list of functions on one finite probability space, with $Z_\ell \in \mathbb{F}_2$ and G_ℓ finite. Suppose $0 \le R < \min_\ell H(Z_\ell \mid G_\ell)$. Take n independent copies of this joint space and let $k = \lfloor nR \rfloor$. For a binary $k \times n$ matrix M put*

$$\Delta_\ell(M) = k - H(MZ_\ell^n \mid G_\ell^n) \ge 0,$$

where entropy is evaluated at the fixed matrix M. There is a constant C, independent of n, such that for every n some matrix M_n satisfies $\sum_\ell \Delta_\ell(M_n) \le C$.

The pairs in the list may be arbitrarily dependent. A small normalised deficit means that the hashed bit sequence retains nearly its full entropy per source copy, even with the complete side information. A random binary matrix makes two distinct strings collide with probability 2^{-k}. Jensen's inequality and this collision identity bound the expected deficit; a conditional typicality estimate makes the bound independent of n. Summing the nonnegative deficits selects one matrix for the whole list. Using the same matrix is essential because unrelated hashes would not preserve the parity identity. The proof in Appendix A follows the bounded-deficit route used in Lean.

Lemma 4 (Conditional entropy profiles)**.** *The almost-entropic region is a closed convex cone. If X_A, X_B, X_C, X_D, W are finite random variables, then $S \mapsto H(X_S \mid W)$ belongs to $\overline{\Gamma}_4^*$.*

Proof. The cone property is classical [3, Theorem 1]; we recall the argument needed here. Independent products add entropy vectors. For an entropic vector f and $\alpha > 0$, take an integer $m > \alpha$, m independent realisations of f, and an independent flag with probability α/m of being one. At each coordinate retain the flag and, when it is one, all m samples; otherwise retain a constant. Each nonempty subset entropy is $h_2(\alpha/m) + \alpha f(S)$, which tends to $\alpha f(S)$. Approximation and closedness extend addition and scaling to the closure, giving convexity. Finally

$$H(X_S \mid W) = \sum_{w:\Pr(W=w)>0} \Pr(W = w) H(X_S \mid W = w)$$

is a finite convex combination of entropy vectors of conditional laws. This also explains the contraction operation of [2, Lemma 1]. □

C Accounting for all fifteen entropy losses

Proof of Theorem 2. Null atoms may be discarded. If $c = 0$ only $R = 0$ is at issue; otherwise fix $0 < R < c$ and $k = \lfloor nR \rfloor$. Form every block variable from the same n independent copies of the original four-variable source. Apply Lemma 3 to the six pairs (L_{ij}, X_iX_j) and the four label–singleton pairs in (6). Select one common matrix $M = M_n$. Define

$$K_i = MY_i^n, \qquad W = (K_A, K_B, K_C),$$

and the nonnegative deficits

$$\begin{aligned} \Delta_{ij} &= k - H(ML_{ij}^n \mid X_i^n X_j^n), \\ \delta_i &= k - H(K_{j(i)} \mid X_i^n), \qquad j(A) = B,\ j(B) = A,\ j(C) = D,\ j(D) = C, \\ D_n &= \sum_{ij} \Delta_{ij} + \sum_i \delta_i \le C. \end{aligned} \tag{7}$$

Here ij ranges over unordered pairs. The common map gives $K_D = K_A + K_B + K_C$ exactly. For every pair ij, basis completion is an invertible change of coordinates between W and (K_i, K_j, ML_{ij}^n). Since K_i, K_j are known from $X_i^n X_j^n$, this gives the exact identity

$$H(W \mid X_i^n X_j^n) = H(ML_{ij}^n \mid X_i^n X_j^n) = k - \Delta_{ij}. \tag{8}$$

For a singleton i, use its partner $j = j(i)$ in (7). The same basis change and the chain rule give

$$\begin{aligned} H(W \mid X_i^n) &= H(K_j \mid X_i^n) + H(ML_{ij}^n \mid X_i^n, K_j) \\ &\geq k - \delta_i + H(ML_{ij}^n \mid X_i^n, X_j^n) = 2k - \delta_i - \Delta_{ij}. \end{aligned}$$

The inequality holds because K_j is a function of X_j^n. The upper bound is $2k$, since K_i is already known. Any triple determines W, so larger observed sets leave no uncertainty.

To bound $H(W)$ itself, use the invertible change $W \leftrightarrow (K_A, K_B, ME^n)$ and condition each chain-rule term on the corresponding full observations:

$$\begin{aligned} H(W) &= H(K_A) + H(K_B \mid K_A) + H(ME^n \mid K_A, K_B) \\ &\geq H(K_A \mid X_B^n) + H(K_B \mid X_A^n) + H(ME^n \mid X_A^n X_B^n) \\ &= 3k - \delta_B - \delta_A - \Delta_{AB}. \end{aligned}$$

Also $H(W) \leq 3k$. Thus $H(W) = 3k - d_0$ and $H(W \mid X_i^n) = 2k - d_i$, with $0 \leq d_0, d_i \leq D_n$. The complete entropy accounting is

Observed subset size	1	2	3 or 4
$H(W \mid X_S^n)$	$2k - d_i$	$k - \Delta_{ij}$	0
$H(W) - H(W \mid X_S^n)$	$k + O(1)$	$2k + O(1)$	$3k + O(1)$

Since $H(X_S^n) = nh(S)$ by independence of the copies, the identity

$$H(X_S^n \mid W) = nh(S) - H(W) + H(W \mid X_S^n)$$

now proves, for all fifteen nonempty subsets,

$$\big|H(X_S^n \mid W) - \big(nh(S) - ku(S)\big)\big| \leq D_n. \tag{9}$$

The errors are $d_0 - d_i$ for singletons, $d_0 - \Delta_{ij}$ for pairs, and d_0 for larger sets; each absolute value is at most D_n.

Passing to an almost-entropic vector. For each finite n, let $p_w = \Pr(W = w)$ and let h_w be the entropy vector of the four blocks conditioned on $W = w$. The conditional profile is the finite average

$$\bar{h}_n(S) = H(X_S^n \mid W) = \sum_{w : p_w > 0} p_w h_w(S).$$

Lemma 4 gives $\bar{h}_n/n \in \overline{\Gamma_4^*}$. Dividing (9) by n, with $k/n \to R$ and $D_n/n \to 0$, gives $\bar{h}_n/n \to h - Ru$ coordinatewise. Closedness proves membership for $0 < R < c$, then for $R = c$ by taking $R \uparrow c$. The case $R = 0$ is immediate. This passage uses an average of conditional profiles; a single conditioning value is not required to realise that average. □

III The source and its parity structure

To generate one source sample, first choose a column with probabilities (P, Q, T), then choose one of its eight rows uniformly. Read the four symbols as the observations of A, B, C, D. Thus the finite source is completely specified by three weights and the table.

Here $P, Q, T > 0$ sum to one. Each column of Table 2 lists eight atoms in $\{0, 1, 2, 3\}^4$, each of probability $P/8$, $Q/8$ or $T/8$. All other probabilities vanish. This is exactly the $s = t = 0$ specialisation of the published family [2, Example 2], with $(p, q, r) = (P, Q, T)/8$; their $\xi_i, \xi_j, \xi_k, \xi_l$ are our A, B, C, D, and their score $\mathbb{I}_{ij}$ is our $\mathrm{J}(h)/h(N)$. We use

$$(P, Q, T) = \frac{1}{100000000}(74830195, 22122546, 3047259). \tag{10}$$

probability $P/8$	probability $Q/8$	probability $T/8$
0000	0210	0011
0101	0321	0120
1010	1100	1002
1212	1332	1230
2121	2001	2103
2323	2233	2331
3232	3012	3213
3333	3123	3322

Table 2: The 24-atom source, in the coordinate order A, B, C, D. Each of the eight atoms in a column has its column's displayed probability.

Subset	Marginal masses	Masses after adding the missing form
A, B, C, D	$4 \times \frac{1}{4}$	—
AB	$8 \times \frac{Q}{8}$, $8 \times \frac{P+T}{8}$	the 24 source masses
AC, BD	$4 \times \frac{P}{4}$, $8 \times \frac{Q+T}{8}$	the 24 source masses
AD, BC	$8 \times \frac{1}{8}$	$8 \times \frac{P}{8}$, $8 \times \frac{Q+T}{8}$
CD	$8 \times \frac{T}{8}$, $8 \times \frac{P+Q}{8}$	the 24 source masses
$\lvert S\rvert \geq 3$	the 24 source masses	—

Table 3: Mass multiplicities; $r \times m$ denotes r masses of size m. "The 24 source masses" means that the observation distinguishes all atoms. The missing forms are those in (14). Every displayed list sums to one.

Write

$$\mathcal{H} = -P\log P - Q\log Q - T\log T, \qquad h_2(x) = -x\log x - (1-x)\log(1-x).$$

Every singleton is uniform on four symbols, and every triple distinguishes all 24 atoms. Counting the masses in each marginal gives all fifteen coordinates:

S	$h(S)$
A, B, C, D	2
AB	$3 + h_2(Q)$
AC, BD	$3 - P + h_2(P)$
AD, BC	3
CD	$3 + h_2(T)$
$\lvert S\rvert \geq 3$	$3 + \mathcal{H}$.

(11)

Table 3 makes these counts and the later completion entropies directly checkable.

In particular the source is tight, i.e. $h(N \setminus \{i\}) = h(N)$ for every i, and

$$b = \mathcal{H} - h_2(Q) = (P+T)h_2\Big(\frac{T}{P+T}\Big) > 0. \tag{12}$$

Tightness gives $I(C; D \mid AB) = h(N) - h(AB) = b$. The expression for b has a direct interpretation in the table: AB identifies the Q-column cases, while the P and T columns have the same eight AB values. In those ambiguous cases, of total probability $P+T$, the unresolved column choice has probabilities $P/(P+T)$ and $T/(P+T)$. The bit $E = \chi(A) + \chi(C)$ defined below resolves this choice, so $H(E \mid AB) = H(CD \mid AB) = b$. This is the randomness that limits the subtraction rate.

The structure needed for hashing appears after a simple local relabelling. Define $\chi(0) = \chi(3) = 0$, $\chi(1) = \chi(2) = 1$ and put

$$U = \chi(A), \quad V = \chi(B), \quad E = \chi(A) + \chi(C),$$

where bit addition is in $\mathbb{F}_2$. Directly on every atom,

$$(\chi(A), \chi(B), \chi(C), \chi(D)) = (U, V, U + E, V + E). \tag{13}$$

These are precisely the forms in (5), so the parity hypothesis of Theorem 2 holds. Their algebraic ranks do not imply probabilistic independence: for example, $H(\chi(A), \chi(C)) = 1 + h_2(P) < 2$. The additional s- and t-columns in the original family have odd label parity under this χ, which is why we set $s = t = 0$. The rational weights were selected by numerical exploration of the new score; their optimality is not used.

Each pair of observed variables reveals two of the three underlying binary forms. The remaining uncertainty sets the rate at which a third form can be hashed privately from that pair. For each pair choose a missing form L_{ij} completing its two observed rows to a basis. Its conditional entropy is as follows:

ij	L_{ij}	$H(L_{ij} \mid X_iX_j)$
AB	E	$\mathcal{H} - h_2(Q) = b$
AC	V	$\mathcal{H} + P - h_2(P)$
AD	V	$h_2(P)$
BC	U	$h_2(P)$
BD	U	$\mathcal{H} + P - h_2(P)$
CD	U	$\mathcal{H} - h_2(T)$.

(14)

Here $(AB, E), (AC, V), (BD, U), (CD, U)$ each distinguish all 24 atoms, giving four entries by subtraction in (11). The completed AD and BC marginals have eight masses $P/8$ and eight masses $(Q + T)/8$, so their joint entropy is $3 + h_2(P)$, giving the other two entries.

Every entry of (14) is at least b, with no numerical comparison required; we use only $0 < T < Q < 1/2$. For CD the entry minus b is $h_2(Q) - h_2(T) \geq 0$. For AC and BD it is $P + h_2(Q) - h_2(P)$, and $h_2(Q) \geq 2Q$ by concavity on $[0, 1/2]$ while $h_2(P) \leq 1$, whence

$$P + h_2(Q) - h_2(P) \geq P + 2Q - 1 = Q - T > 0.$$

For AD and BC: on the P-column atoms $A = C$, while $\chi(A) \neq \chi(C)$ on the other sixteen atoms. So $E = 0$ exactly on the P-column, $\Pr(E = 0) = P$, and $b = H(E \mid AB) \leq H(E) = h_2(P)$. Hence every entry is at least b, and the AB entry equals b. Theorem 2 therefore gives $h - Ru \in \overline{\Gamma_4^*}$ for $0 \leq R \leq b$, proving the first assertion of Theorem 1.

IV The achieved score

The source coordinates give

$$I(C; D \mid A) = I(C; D \mid B) = 1 - P + h_2(P) - \mathcal{H},$$
$$I(A; B) = 1 - h_2(Q), \qquad I(C; D) = 1 - h_2(T).$$

The violation comes from a large $I(C; D)$ relative to the two conditional mutual informations and $I(A; B)$. Since $\mathrm{J}(u) = 0$, we obtain

$$\mathrm{J}(z) = \mathrm{J}(h) = 2 - 2P + 2h_2(P) + h_2(T) - 2\mathcal{H} - h_2(Q), \tag{15}$$
$$z(N) = 3 + \mathcal{H} - 3b = 3 - 2\mathcal{H} + 3h_2(Q). \tag{16}$$

For a tight source the Matúš–Csirmaz transformation preserves $\mathrm{J}(h)$ and reduces the joint entropy from $h(N)$ to $h(AB) = h(N) - b$ [2, Theorem 4 and Section 8]. The parity construction reduces it

to $h(N) - 3b$. At these fixed weights the signed scores are

$$\sigma(h) = -0.08365470562\ldots, \qquad \frac{\mathrm{J}(h)}{h(AB)} = -0.08778073130\ldots, \qquad \sigma(z) = -0.09738743692\ldots.$$

Thus this comparison isolates the gain from the new transformation.

All target coordinates are specified exactly by $z(S) = h(S) - b\min\{3, |S|\}$; Table 4 lists their numerical values, grouping only coordinates with equal values. In particular,

Subset(s) S	$z(S)$ (bits)	loss $h(S) - z(S)$
A, B, C, D	1.814430591039	b
AB	3.391260034510	$2b$
AC, BD	2.694517625911	$2b$
AD, BC	2.628861182078	$2b$
CD	2.825617546930	$2b$
$ABC, ABD, ACD, BCD, ABCD$	3.391260034510	$3b$

Table 4: All fifteen coordinates of the limiting almost-entropic point.

$$b = 0.18556940896122\ldots, \quad \mathrm{J}(z) = -0.33026612269970\ldots, \quad z(N) = 3.39126003450969\ldots > 0.$$

Appendix B gives a rational logarithm-series calculation that certifies (4). The independent exact checker also accumulates all fifteen marginals directly from the source atoms. The enclosure evaluates the exact expressions above; it is separate from the almost-entropic membership argument.

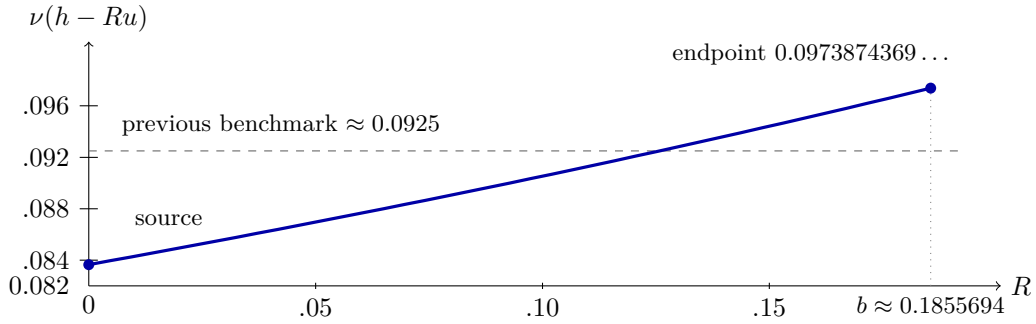


Figure 1: The achieved violation on the fixed path $h - Ru$, $0 \le R \le b$. The endpoint is the Shannon limit for this direction, proved achievable in the almost-entropic cone. The vertical axis begins at 0.082. The horizontal line is the rounded previous benchmark. This is a plot of the construction, not a section of the unknown cone.

Geometry along this path. The function u contributes $u(ABC)+u(ABD)-u(AB)-u(N) = 1$ to $I(C; D \mid AB)$, so

$$I_{h-Ru}(C; D \mid AB) = b - R.$$

The privacy bottleneck is therefore also the Shannon obstruction on this path: every $R > b$ violates submodularity, while Theorem 1 realises every $0 \le R \le b$. The nonnegative feasible rates on this path are exactly $[0, b]$, both for $\overline{\Gamma_4^*}$ and for Γ_4. At the endpoint, $z(AB) = z(N)$ and every triple also has entropy $z(N)$, placing $z/z(N)$ on these Shannon boundary hyperplanes. Since $\mathrm{J}(h - Ru) = \mathrm{J}(h) < 0$, its violation $-\mathrm{J}(h)/(h(N) - 3R)$ increases along the segment and is largest at $R = b$. This proves optimality along this fixed source and direction.

Finally, since $z \in \overline{\Gamma_4^*}$, finite entropic vectors approach z. The score is continuous at z because $z(N) > 0$. Its upper bound in (4) is strictly below -0.09738, proving the final assertion of Theorem 1. Exact finite realisability of z remains open. The next section gives an exact sampling procedure for a suitable finite law, selected using the strict score margin.

V Generating a finite four-variable distribution

The source table is already an explicit small distribution, but its ordinary score is only about -0.08365470562. The score near -0.09738744 belongs to the limiting vector z. We now describe actual finite distributions which exceed the stated threshold, by selecting a conditional law.

Fix the strictly subcritical rational rate

$$R_0 = \frac{371}{2000} = 0.1855 < b.$$

The same exact entropy formulas certify

$$\frac{\mathrm{J}(h - R_0 u)}{(h - R_0 u)(N)} = -0.09738145760031\ldots < -0.09738.$$

For all sufficiently large n, the matrix selected in the proof has a conditional average profile with positive total entropy and score below -0.09738. Let h_w be the entropy vector of the four blocks given $W = w$, and write $p_w = \Pr(W = w)$. Its average is $\bar{h} = \sum_w p_w h_w$. The exact identity

$$\frac{\mathrm{J}(\bar{h})}{\bar{h}(N)} = \sum_{w : h_w(N) > 0} \frac{p_w h_w(N)}{\bar{h}(N)} \frac{\mathrm{J}(h_w)}{h_w(N)}$$

shows that at least one positive-entropy fibre has score no larger than the average. Fibres with $h_w(N) = 0$ have all subset entropies and $\mathrm{J}(h_w)$ equal to zero. This selects a fibre for the scalar score; it does not make its fifteen coordinates equal to the average profile.

Once such n, $k = \lfloor nR_0 \rfloor$, M and w have been chosen, the following is an exact sampling recipe.

1. Independently at each of n positions, choose a column with probabilities (P, Q, T) and a uniform row among the eight table rows. Read off $(X_{A,t}, X_{B,t}, X_{C,t}, X_{D,t})$.
2. Compute $K_i = M(\chi(X_{i,1}), \ldots, \chi(X_{i,n}))^{\mathsf{T}}$ over $\mathbb{F}_2$, for $i = A, B, C$, and set $W = (K_A, K_B, K_C)$.
3. Retain the entire trial exactly when $W = w$; otherwise repeat it. Output the four retained words, each in $\{0, 1, 2, 3\}^n$.

This rejection sampler terminates almost surely because $p_w > 0$. For four words $x = (x_A, x_B, x_C, x_D)$, if π is the one-letter rational table law, the output law is exactly

$$\Pr(X'_A = x_A, X'_B = x_B, X'_C = x_C, X'_D = x_D) = \frac{\mathbf{1}\{W(x) = w\}}{p_w} \prod_{t=1}^{n} \pi(x_{A,t}, x_{B,t}, x_{C,t}, x_{D,t}).$$

In particular all its probabilities are rational.

The proof guarantees suitable finite choices, but a practical block length, matrix and fibre have not been computed. The strict rate $R_0 < b$ leaves a small privacy margin; no efficiency bound is claimed.

VI Formal verification

Theorem 1 has been formalised in Lean 4 (version 4.35.0-rc2), using Mathlib commit `065356127b1d`, and checked by the Lean kernel. The declaration is `ConditionalParityHashing.certified_main` in `CertifiedMain.lean`. In the notation of this paper it states:

(i) $h - Ru \in \overline{\Gamma_4^*}$ for every $0 \le R \le b$;
(ii) $z(N) > 0$;
(iii) $-97387436923/10^{12} \le \mathrm{J}(z)/z(N) \le -97387436922/10^{12}$;
(iv) there exist an alphabet size m and a law on $\{0, \ldots, m-1\}^4$ whose entropy vector f has $f(N) > 0$ and $\mathrm{J}(f)/f(N) < -0.09738$.

The parameter in the final existential quantifier is an alphabet size, rather than the block length used in the construction.

The definitions in this statement have their ordinary mathematical meanings. `sourceLaw` is the rational law of Table 2. `finiteEntropy` computes Shannon entropy in bits from the finite-law

marginals. `almostEntropicProfiles` is the topological closure of those entropy vectors, in the space of functions on all sixteen subsets of N, including the zero empty-set coordinate. The checked semantic bridge identifies membership with simultaneous ε-approximation of all coordinates by an actual finite law. `paperJ` is precisely (1); `criticalRate` is $h(N) - h(AB)$, proved equal to $I(C; D \mid AB)$; `shiftedProfile` R is $h - Ru$; and `candidate` is z. The general six-test criterion is also checked, through the ten-threshold argument and its six-to-ten reduction. The axiom report uses only `propext`, `Classical.choice` and `Quot.sound`.

The formal proof uses the same bounded-deficit hashing argument as Appendix A. Its source-threshold comparisons and score bounds use certified 20-term artanh-series enclosures, while the written threshold comparisons are analytic and the independent Python checker uses the rational series in Appendix B. The finite-alphabet clause follows formally by density and continuity; the explicit fibre selection and sampling description of Section V, and the path-optimality discussion, are not separately formalised. The proof package contains the pinned Lean sources (27 proof modules and one verification file), a fresh-directory rebuild script with the axiom report it produced, and two exact rational checkers written for the Python standard library. The paper, manuscript sources, Lean proof and verification evidence are available in the GitHub repository. The archived version is identified by doi:10.5281/zenodo.23662511. The paper and its sources are licensed under CC BY 4.0; code is licensed under Apache-2.0.

VII Discussion

The construction gives an inner bound for the full almost-entropic cone. The maximum Ingleton violation, extremality of z, and exact finite realisability of z remain open. The six-test criterion suggests looking for other finite sources with local algebraic relations that permit larger structured subtractions. Extending the construction to other matroidal information profiles may provide further inner bounds.

A Proof of the simultaneous hashing lemma

Proof. Fix one pair (Z, G) in the finite list, put $H = H(Z \mid G)$ and $\varepsilon = (H - R)/2 > 0$, and take $n \geq 1$. For a positive-probability value g of G^n, write $p(z \mid g) = \Pr(Z^n = z \mid G^n = g)$. Call (z, g) typical when $p(z \mid g) \leq 2^{-n(H-\varepsilon)}$, and let a_n be the probability of the typical event. All logarithms in this proof are to base two.

For a fixed binary matrix M define

$$q_M(z, g) = \Pr(MZ^n = Mz \mid G^n = g).$$

Thus $H(MZ^n \mid G^n) = \mathbb{E}[-\log q_M(Z^n, G^n)]$ is a sum of nonnegative terms. Drop the atypical terms and average over a uniform random matrix M, independently of the source. If $a_n > 0$, Jensen's inequality gives

$$\mathbb{E}_M H(MZ^n \mid G^n) \geq -a_n \log \mathbb{E}\big[\mathbb{E}_M q_M(Z^n, G^n) \mid \text{typical}\big].$$

Two distinct binary strings collide under a uniform $k \times n$ matrix with probability 2^{-k}: each row has a uniform dot product with their nonzero difference, and the rows are independent. Consequently

$$\mathbb{E}_M q_M(z, g) = \sum_{z'} p(z' \mid g) \Pr_M(Mz' = Mz) \leq p(z \mid g) + 2^{-k}.$$

On typical pairs the right-hand side is at most $2^{-k}(1 + 2^{k-n(H-\varepsilon)})$. Write $x = 2^{k-n(H-\varepsilon)}$. Since $\log_2(1 + x) \leq x/\ln 2$ and $a_n \leq 1$, the preceding estimate gives

$$\mathbb{E}_M H(MZ^n \mid G^n) \geq a_n\big(k - \log_2(1 + x)\big) \geq a_n k - \frac{x}{\ln 2}.$$

Subtracting from k and using $k \le nR = n(H - 2\varepsilon)$ gives

$$\mathbb{E}_M\,[k - H(MZ^n \mid G^n)] \le (1 - a_n)k + \frac{2^{k-n(H-\varepsilon)}}{\ln 2} \le (1 - a_n)nR + \frac{2^{-n\varepsilon}}{\ln 2}. \tag{17}$$

If $a_n = 0$, the first bound holds directly from nonnegativity of entropy, so no conditioning on a null event is required.

On the finite positive support, $\iota = -\log p(Z \mid G)$ is bounded, with mean H and a finite variance v. For the tth copy put $\iota_t = -\log p(Z_t \mid G_t)$. Independence gives $-\log p(Z^n \mid G^n) = \sum_{t=1}^{n} \iota_t$. Chebyshev's inequality therefore yields $1 - a_n \le v/(n\varepsilon^2)$. It follows that

$$\mathbb{E}_M\,[k - H(MZ^n \mid G^n)] \le \frac{vR}{\varepsilon^2} + \frac{1}{\ln 2}.$$

Sum this bound over the finite list. Its sum is a constant independent of n, and some deterministic matrix has a deficit sum no larger than the average. For $n = 0$ every key is empty and every deficit is zero. □

B Exact numerical enclosure

The following rational computation certifies (4); it does not enter the membership proof. For a rational $x \in [1, 2]$ put $y = (x - 1)/(x + 1)$ and

$$\ell_m(x) = 2\sum_{j=0}^{m-1} \frac{y^{2j+1}}{2j+1}, \qquad 0 \le \ln x - \ell_m(x) \le \frac{2y^{2m+1}}{(2m+1)(1-y^2)}. \tag{18}$$

This is the positive series for $2\operatorname{arctanh} y$, with its tail bounded geometrically. For each positive rational argument w, write $w = 2^{\ell}x$, $1 \le x < 2$, and enclose $\log_2 w = \ell + \ln x/\ln 2$ using (18) with $m = 42$ for x and 2. Rational interval arithmetic in (12), (15) and (16) gives (4). The accompanying `verify_paper.py` implements exactly this recipe using only rational arithmetic from the Python standard library and also checks the finite source identities. The displayed decimal bounds are outward rounded; the exact entropy expressions define the point.

The Lean proof independently encloses the same logarithms with Mathlib's 20-term artanh-series bounds. Both calculations are rigorous.

Acknowledgements and contributions

Søren Riis is the creator and coordinator of the project, responsible for its research direction and outline. Mathematical development, computation and verification were carried out with Astra/Codex and Claude; Claude also contributed the Lean formalisation and adversarial review, and Astra/Codex checked the formal statements and rebuilt the proof package. These AI-assisted checks do not constitute independent human peer review.

The source family is due to Matúš and Csirmaz; the compression, privacy-amplification and conditioning methods build on the works cited above. The work of Dougherty, Freiling, Zeger and Yeung supplies much of the surrounding information-inequality framework [1, 3, 6]. These attributions do not imply endorsement by the researchers named.

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